

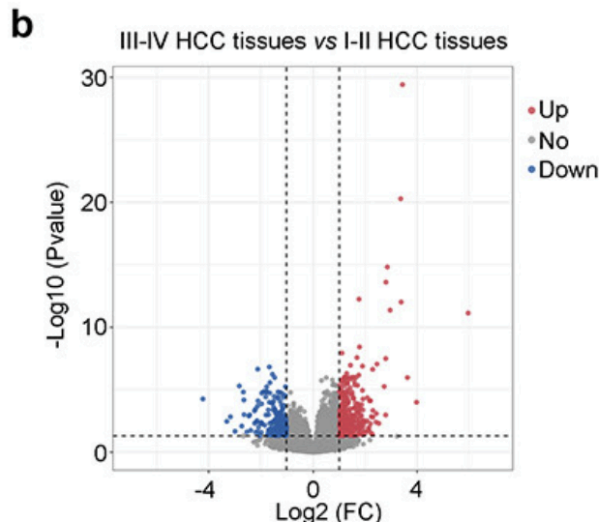
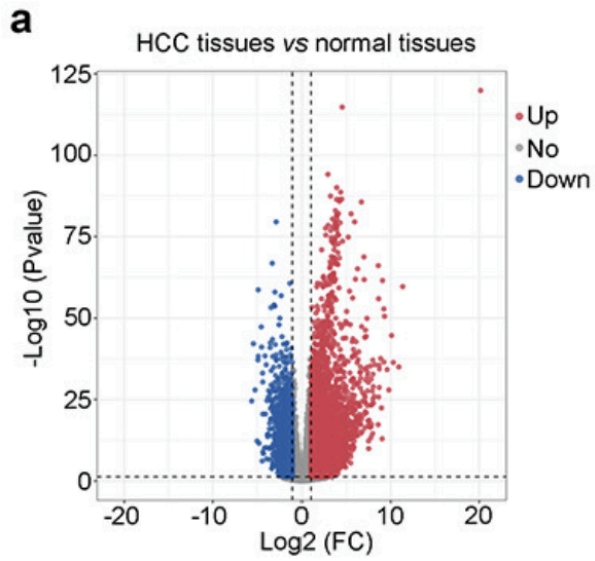
Module 6: Differential gene analysis

Key Concepts

- Differential gene analysis
- Pre-processing RNASeq data
 - Read count normalization and transformation
 - Gene filtering strategies
- Gene Set Enrichment Analysis
- Over-representation Analysis

Differential gene analysis

- The aim of a differential gene analysis is to identify genes with difference in expression patterns under different experimental conditions (gene(s) versus condition(s) and vice versa) for a set of samples.
- By asking “which genes change and by how much?” and then mapping those changes to **pathways, processes, and regulatory mechanisms**, we can generate hypotheses about the underlying biological mechanisms.
 - **Disease mechanisms** (e.g., identifying driver genes in cancer)
 - **Therapeutic targets** or potential drug mechanisms
 - **Biomarker discovery** for diagnosis and prognosis
 - **Fundamental biology** of development, stress responses, or adaptation.



Which genes are up- or down-regulated in a given condition?

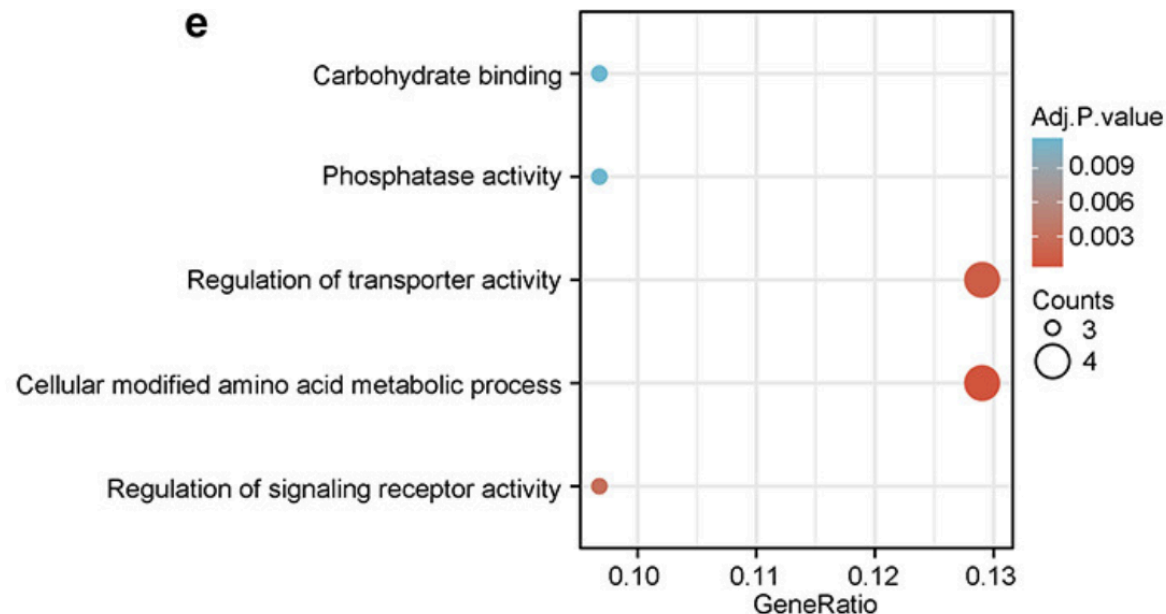
Question: Under condition A vs. condition B (e.g., treated vs. untreated), which genes show significant changes in expression levels?

- The treatment can be drug exposure, knock-out mutation, or any external stimulus
- Identifying up-regulated genes can point toward processes or pathways that are activated under specific conditions, while down-regulated genes might indicate suppression or reduced activity of certain pathways.

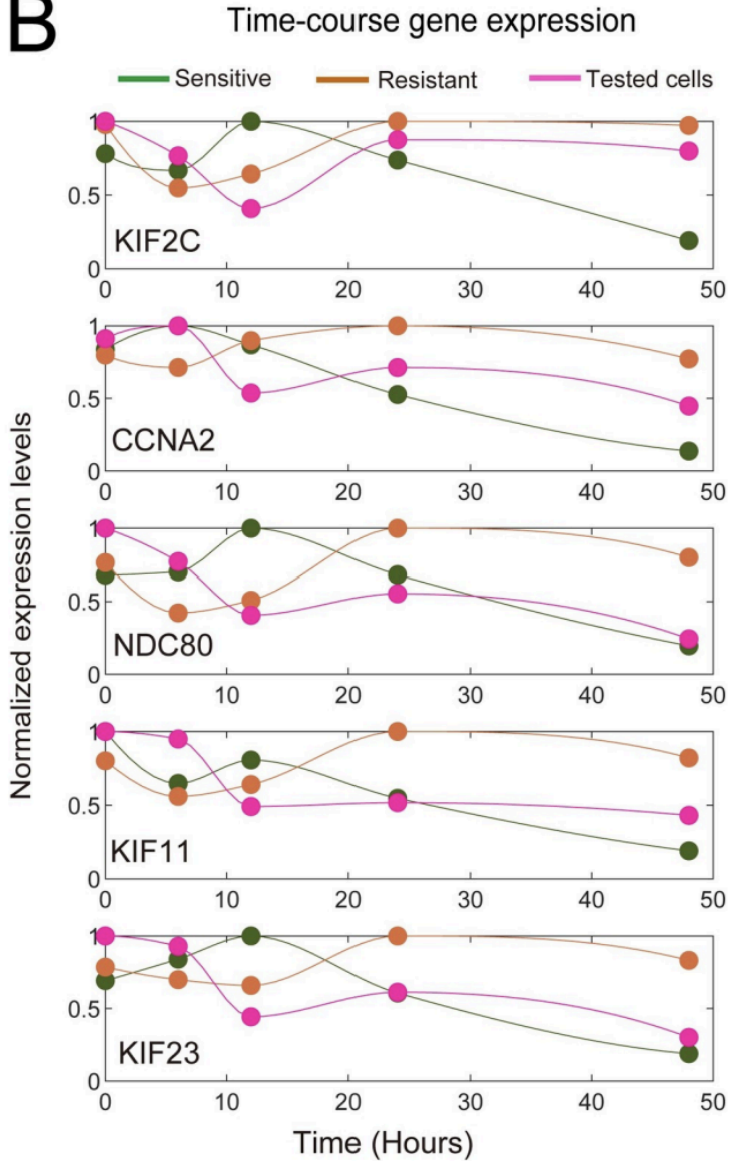
What pathways or biological processes are affected?

Question: Are the significantly differentially expressed genes enriched in specific functional categories, pathways, or Gene Ontology terms?

- DGE results often feed into **gene set enrichment analysis (GSEA)** or pathway analysis (gProfiler) to reveal broader biological programs at play—like inflammation, cell cycle regulation, or metabolic shifts.



B

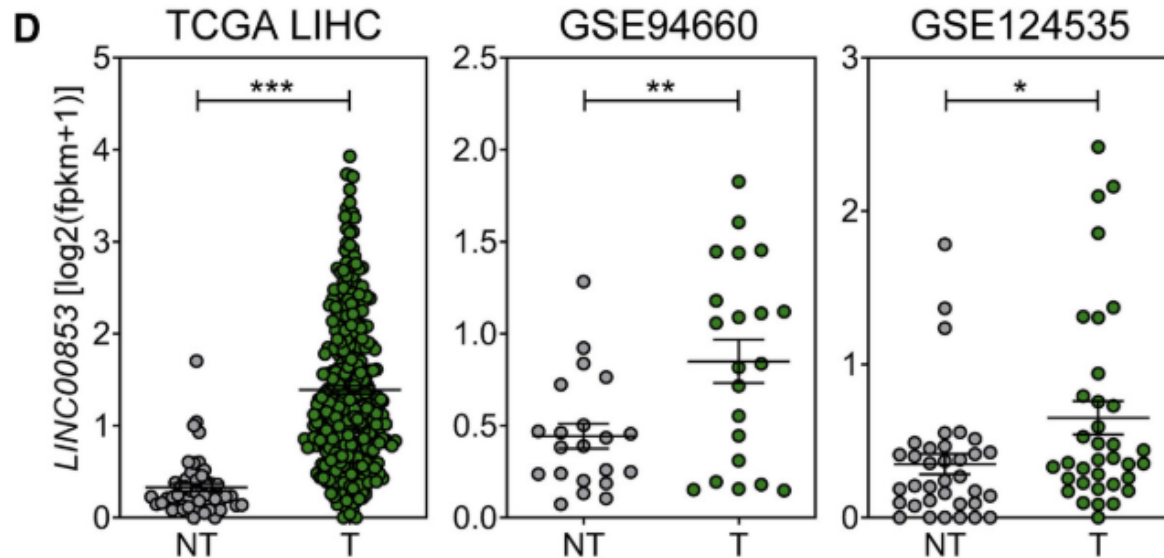


How does gene expression change over time (time-course experiments)?

- Question: What genes are differentially expressed at early vs. late stages of a response, disease progression, or development?
 - Time-course DGE studies can reveal **dynamic gene expression trends** that can help identify early-response genes, late-stage markers, or transitions in biological processes.

Are there biomarker candidates for disease or treatment response?

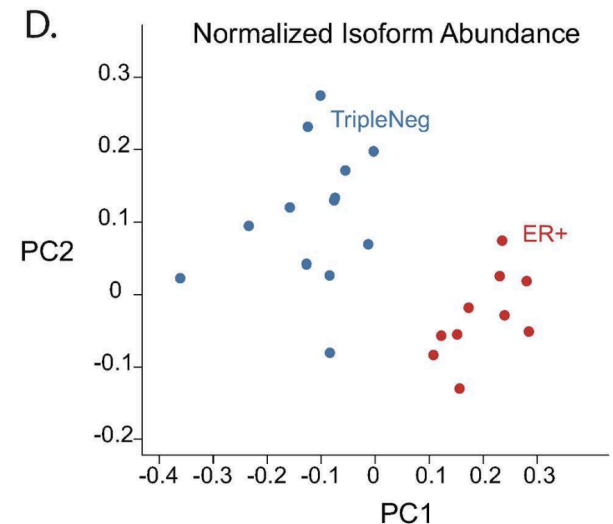
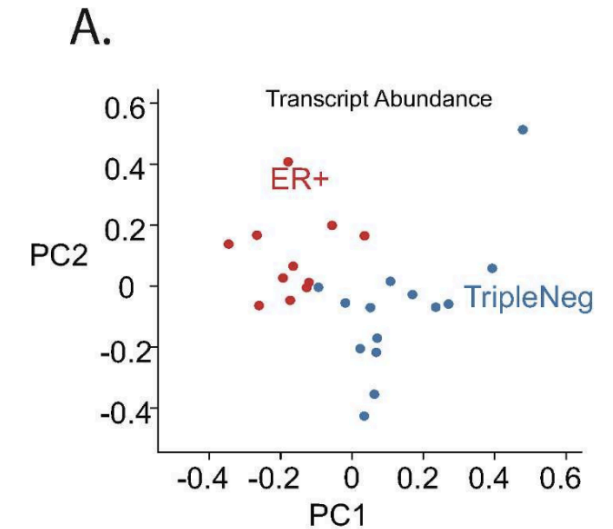
- **Question:** Which genes consistently show differential expression in a disease state (e.g., tumor vs. normal tissue) or in response to a therapy?
 - Such genes may serve as **potential biomarkers** for diagnostic, prognostic, or therapeutic monitoring purposes (e.g., predicting drug resistance or disease recurrence).



LINC00853 expression in the nontumor and the HCC cohorts in three HCC RNA-Seq datasets (TCGA_LIHC, GSE94660, and GSE124535).

Can we classify or subtype biological samples based on gene expression profiles?

- **Question:** In a cohort of samples (e.g., different cancer subtypes), which sets of genes distinguish one subtype from another?
 - Identifying **signature genes** for subtypes can guide more personalized treatment strategies
- **Example:** Different subtypes of breast cancer have very different expression profiles. These expression profiles are associated with different patient outcomes and with different treatment protocols. Different subtypes of breast cancer can be distinguished by using differential transcript isoform expression.



DGE Tools

Tool	Statistical Model	Key Strengths	When to Use
DESeq2	- Negative binomial with empirical Bayesian shrinkage of dispersion and fold changes - Based on the hypothesis that most genes are not DE	- Better handling of overdispersion - Good documentation and built-in plotting (MA plots, PCA) - Shrinkage of fold changes helps stabilize LFC for low-count genes	- Moderate sample sizes (≥ 3 replicates/condition) - Typical RNA-seq designs (two-group or multi-factor) - Ideal if you want an easy-to-follow pipeline with built-in diagnostics

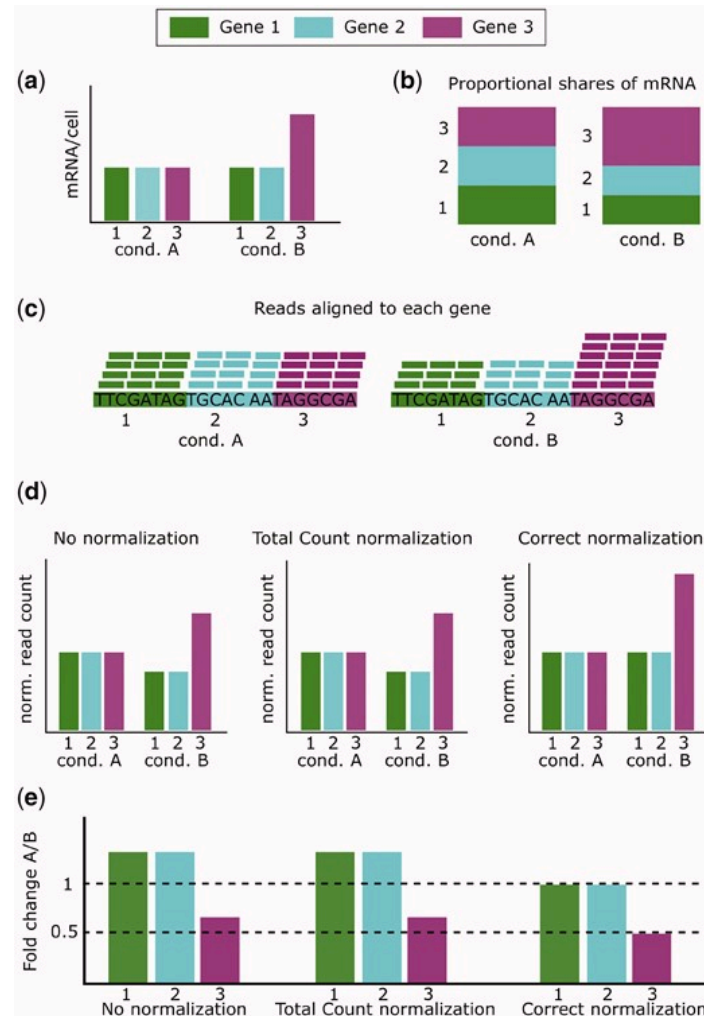
DGE Tools

Tool	Statistical Model	Key Strengths	When to Use
edgeR	- Negative binomial with empirical Bayes estimation of dispersion - Based on the hypothesis that most genes are not DE	- Highly flexible for complex or unbalanced designs - Known to perform robustly even with smaller sample sizes - Deep configurability for normalization, dispersion, and model setup	- Complex experimental designs (time series, multi-factor) - Smaller datasets (though more replicates is still better) - Prefer a “light-weight” approach with fine-grained control

DGE Tools

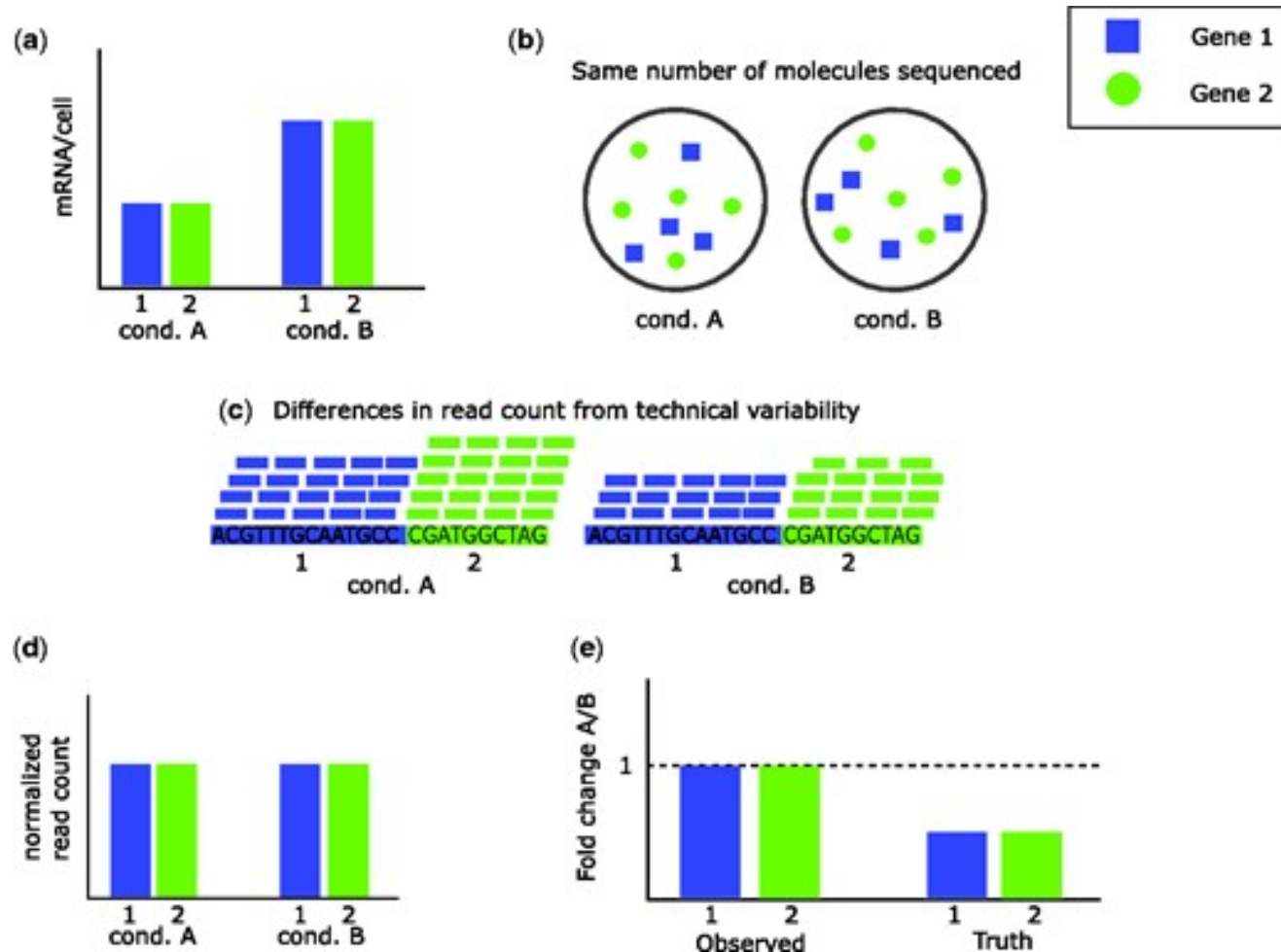
Tool	Statistical Model	Key Strengths	When to Use
limma (with voom)	Linear models with empirical Bayes shrinkage of variance estimates (after voom transformation)	- Stable release (works with microarray data) - Good for large datasets and complex designs - Fast and memory-efficient; integrates well with other limma functions (e.g., contrasts, gene set testing)	- Already familiar with limma or microarray workflows - Large sample sizes or multi-factor, repeated-measures designs - Looking for flexible linear modeling framework and advanced downstream analyses (e.g., GSEA, contrasts)

Read Count Normalization: Why do we need to do this?



- The correct normalization method to use depends on which assumptions are valid for the biological experiment.
- Incorrect normalization leads to problems in downstream analysis, such as inflated false positives, that mean results cannot be trusted.
- **Example:** A small number of highly expressed genes creates the appearance that non-DE are DE, but the false DE calls may be corrected by normalizing read counts so that the expression levels of non-DE genes are equivalent.

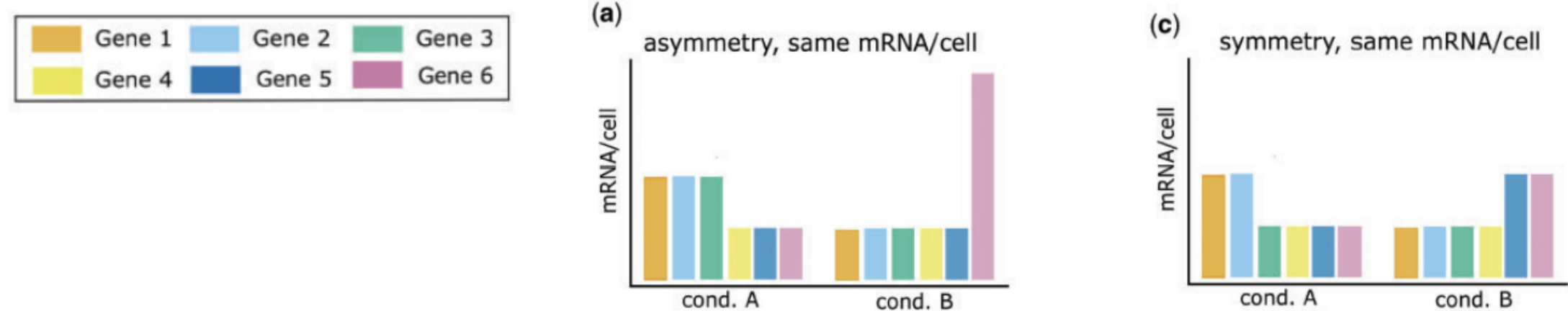
Read Count Normalization: Why do we need to do this?



- In the case of a global shift in expression, it may appear that DE genes are non-DE or that up-regulated genes are down-regulated.
- It is necessary to take into account the differences in overall expression between conditions.

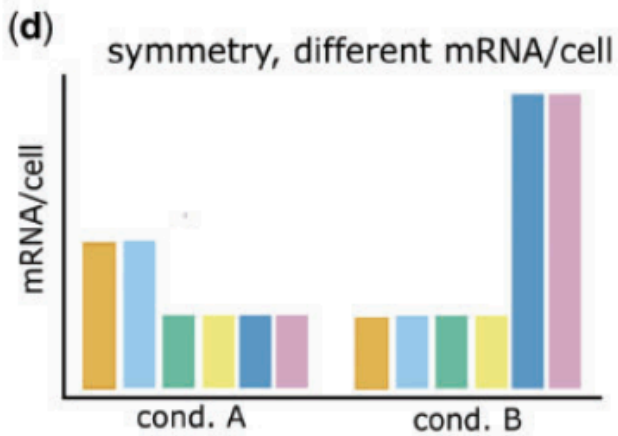
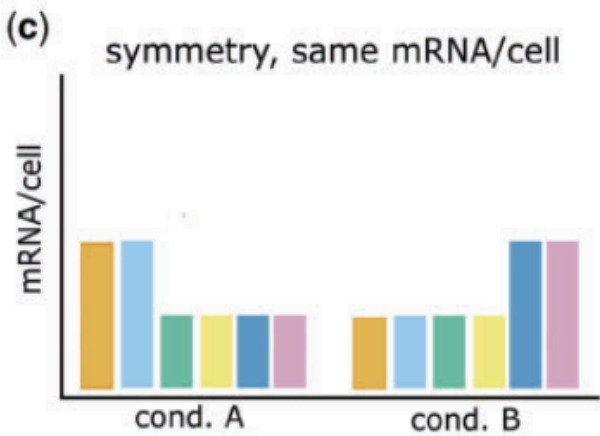
Read Count Normalization: Normalization by Library Size

	Details
Use Case	When the total mRNA remains equal or within the same range across different treatments/conditions, even if there is some asymmetry.
Assumptions	- Total mRNA/cell is consistent across conditions. - Differences in sequencing depth are the primary technical variability.
Limitations	- Cannot handle global shifts in expression (large asymmetry). - Assumes minimal variability in the total mRNA/cell across samples.



Read Count Normalization: Normalization by Distribution

	Details
Use Case	When there is symmetry in sample sizes, even if total mRNA levels differ across conditions.
Assumptions	- DE & non-DE genes behave similarly (technical effects are the same). - Balanced expression: approximately equal numbers of up- and down-regulated genes across conditions.
Advantages	Can handle small asymmetry (e.g., a few highly expressed genes) or differences in total mRNA/cell.
Limitations	Fails to correct global shifts in expression (large asymmetry).

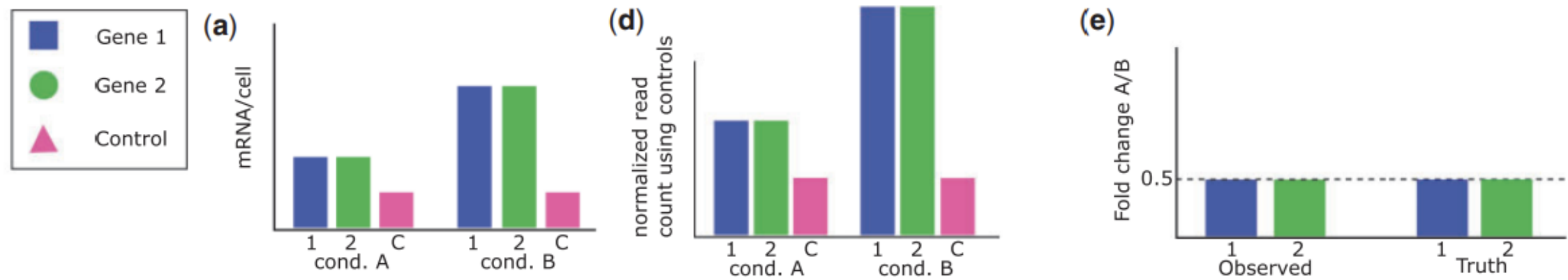


Read Count Normalization: Normalization by Distribution

Method	Key Idea
Quantile Normalization	Forces the distribution of gene expression values (e.g., read counts or log-transformed counts) to match across samples.
Median/Upper Quartile	Normalizes counts based on a specified quantile (e.g., 50th or 75th percentile) rather than the total number of reads.
TMM (Trimmed Mean of M-values)	Trims extreme values (highly expressed genes) and calculates scale factors using average log-fold changes (M-values).
PoissonSeq / DEGES	Iteratively identifies non-differentially expressed (non-DE) genes and estimates scaling factors using those genes only.

Read Count Normalization: Normalization by Controls

	Details
Use Case	When suitable controls exist that remain unaffected by biological conditions and behave consistently across samples.
Assumptions	- Controls exist and are non-DE in the context of the experiment. - Controls reflect technical effects for all genes.
Advantages	Corrects for variations in sequencing depth and technical effects using robust controls.
Limitations	- Requires reliable and representative controls. - Cannot correct global shifts in expression if controls do not reflect all genes.



Read Count Normalization: Normalization by Controls

Method	Key Idea
Using Control Genes	Normalizes data based on a set of genes known (or assumed) to have constant expression across samples (non-DE).
Spike-in Controls	Uses known amounts of external RNA spikes (added to each sample) to determine and correct for technical biases.



	sample A	sample B
gene 1	50	40
gene 2	50	40
...	...	...
gene 99	50	40
gene 100	10	1000

$d_A = 4'960$ $d_B = 4'960$

- **Length bias:** Longer genes are more likely to generate more reads than shorter ones with similar expression levels.
- **Differences in library size composition:** Since the sequencing depth (number of reads per samples) is pre-determined as part of the sequencing strategy, skewed read count distribution within samples can mask finer details in gene expression profile.
- Between-sample differences in library size must be necessarily corrected before comparing samples to detect differentially expressed genes.

Read Count Transformation: Why do we need to do this?

- RNA-seq count data is highly skewed (due to differences in sequencing depth, expression levels, and variance).
- Standard log transformation ($\log_2(\text{count} + 1)$) is not sufficient due to heteroscedasticity (high variance for highly expressed genes), so we use **VST** and **rlog** to help stabilize variance

Feature	Variance Stabilizing Transformation (VST)	Regularized Log Transformation (rlog)
Purpose	Stabilizes variance for high-count genes	Shrinks differences between low-count genes while preserving higher count relationships
Assumes Variance-Mean Relationship?	Yes, variance increases with mean expression	Yes, but applies stronger shrinkage for low counts

Feature	Variance Stabilizing Transformation (VST)	Regularized Log Transformation (rlog)
Dataset size	Optimized for large samples	Computationally expensive for large datasets
Computational Speed	Fast	Slower due to Bayesian shrinkage
Handles Low-Count Genes Well?	Partially, but less aggressive	Yes, applies stronger shrinkage
Normalization	Assumes normalized counts (DESeq2 size factors)	Assumes normalized counts (DESeq2 size factors)
Use Case	Large-scale RNA-seq datasets, differential expression analysis	Small RNA-seq datasets, noise reduction in low-count genes

Gene filtering strategies

Supplemental Slides

Read Count Normalization: What should I use?

- **Use RPKM/FPKM** when comparing gene expression within a **single sample** (e.g., understanding relative expression of genes within the same condition).
- **Use TPM** when comparing gene expression **between samples** because TPM ensures that total expression values are comparable across samples.
- **Use CPM** for **differential expression analysis**, especially with raw counts, as it retains variance information required by statistical methods (e.g., edgeR, DESeq2).

Read Count Normalization

Metric/Method	Name/Description	Calculation	Purpose
Total Count		$(\text{Gene read count}) / (\text{Total reads in sample})$	Adjusts each gene's read count by the total number of reads in the sample.
CPM	Counts Per Million	$(\text{Gene read count} \times 10^6) / (\text{Total mapped reads})$	Normalizes only for sequencing depth (not gene length).
RPKM	Reads Per Kilobase (kb) per Million (single-end)	$(\text{Read count} \times 10^6) / (\text{Total reads} \times \text{Gene length in kb})$	Normalizes for both library size and gene length; often used for single-end RNA-seq.
FPKM	Fragments Per kb per Million (paired-end)	Same as RPKM but accounts for paired-end reads	Normalizes for library size and gene length in paired-end RNA-seq.
		$(\text{Read count} / \text{Gene length in kb}) / (\text{Total reads in sample} / 10^6)$	Normalizes within a sample

Read Count Normalization: Normalization by Library Size

Summary

Overall, **differential gene expression analysis** is essential

DGE thus serves as a gateway to more detailed functional studies, guiding **follow-up experiments** such as knockdowns, overexpression assays, or protein-level validations.

Differential gene analysis What biological questions can we answer?

Tools: DeSeq2, edgeR, limma

Pre-processing data Read Count Normalization RPKM, FPKM, TPM, CPM

Read Count Transformation vst, rlog

Gene filtering strategies

Gene Set Enrichment Analysis What biological questions can we answer?

Statistical measures: Normalized Enrichment Score, FDR, Nominal P-value

Tools: GSEA, ClusterProfiler

Gene set databases: MSigDB, mulea, Enrichr